Author: You are quite right. Sequences (5), (7), (9), (10) and (13) are bounded. Note that (5), (7) and (13) are bounded and at the same time monotonic. Don't you feel that this fact is somewhat puzzling?

Reader: What's puzzling about it?

Author: Consider, for example, sequence (5). Note that each subsequent term is greater than the preceding one. I repeat, each term! But the sequence contains an infinite number of terms. Hence, if we follow the sequence far enough, we shall see as many terms with increased magnitude (compared to the preceding term) as we wish. Nevertheless, these values will never go beyond a certain "boundary", which in this case is unity. Doesn't it puzzle you?

Reader: Well, generally speaking, it does. But I notice that we add to each preceding term an increment which gradually becomes less and less.

Author: Yes, it is true. But this condition is obviously insufficient to make such a sequence bounded. Take, for example, sequence (4). Here again the "increments" added to each term of the sequence gradually decrease; nevertheless, the sequence is not bounded.

↳ V. Imp Counter-argument

Reader: We must conclude, therefore, that in (5) these "increments" diminish faster than in (4).

Author: All the same, you have to agree that it is not immediately clear that these "increments" may decrease at a rate resulting in the boundedness of a sequence.

Reader: Of Course, I agree with that.

Author: The possibility of infinite but bounded sets was not known, for example, to ancient Greeks. Suffice it to recall the famous paradox about Achilles chasing a turtle.

Let us assume that Achilles and the turtle are initially separated by a distance of 1 Km. Achilles moves 10 times faster than the turtle. Ancient Greeks reasoned like this: during the time Achilles covers 1 Km the turtle covers 100 m. By the time Achilles has covered these 100 m, the turtle will have made another 10 m more, and before Achilles has covered these 10 m, the turtle will have made 1 m more, and so on.